

Open Project 2026

Dynamic Tariff Optimization for EV Charging Networks

An agentic AI approach: predict demand, price dynamically, and learn from outcomes

Executive summary

We built a three-agent system that turns flat-rate EV charging into smart, demand-aware pricing.

+2.76%

revenue gain vs ₹15
flat-rate baseline

27%

better than naive
forecast baseline

+2.10%

off-peak demand
uplift from discounts

0.86

R² of demand
prediction model

How it works — a Demand agent forecasts station utilization 2 hours ahead, a Tariff agent surges price above 80% use and discounts below 30%, and a Monitoring agent tracks the outcome each day and adjusts how aggressive the pricing gets.

The data and how we prepared it

Two real-world datasets

ST-EVCDP (Shenzhen) — main dataset

247 stations · 30 days · 5-minute intervals · 2.1M rows after reshaping

ACN-Data (Caltech) — supporting

16,000 real charging sessions, used to study idle-time behaviour

Key preprocessing steps

- Merged 4 wide tables (occupancy, volume, price, duration) into one long table keyed by time + station
- Engineered the core metric: $\text{Utilization} = \text{occupancy} \div \text{total chargers}$
- Built lag features (5 min / 30 min / 1 hr ago) and rolling averages (1 hr, 3 hr)
- Used grid ID (not row index) to join station info — caught and fixed early

Assumptions we made

Baseline price = ₹15/kWh fixed

As specified in the problem statement.

Demand elasticity = -0.3

A 10% price rise reduces demand $\sim 3\%$. A conservative, literature-backed value.

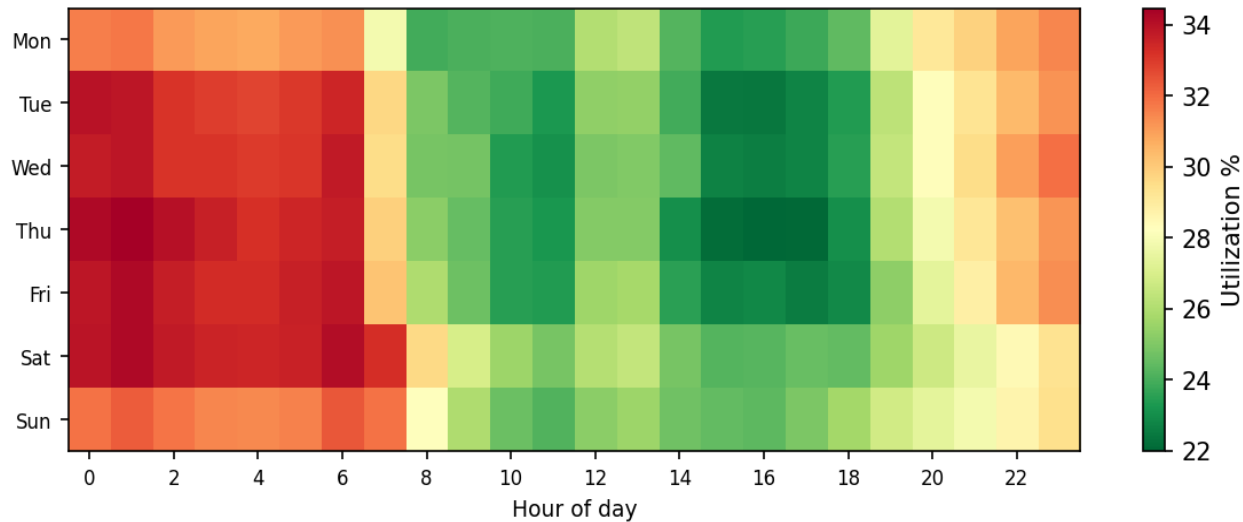
Utilization capped at 0–100%

Sensor glitches occasionally report occupancy above capacity; we clip these.

Missing values dropped at edges only

Lag/target gaps at each station's start and end — about 0.2% of rows.

What the data told us



Utilization by day & hour — the pattern is the same all week

Three findings that shaped our pricing

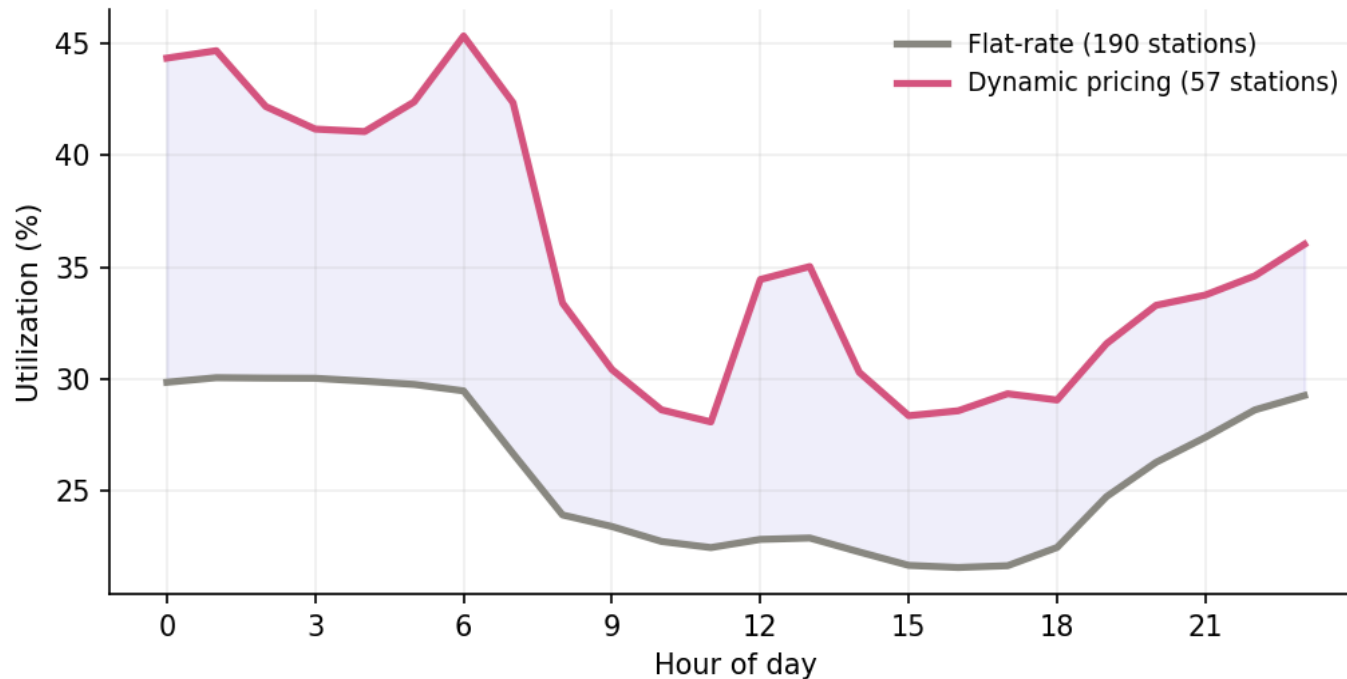
Night is the real peak. Shenzhen stations are busiest 10pm–6am, emptiest mid-afternoon — opposite of the daytime peak you'd expect.

Pattern is highly consistent. Weekday and weekend look nearly identical, so hour-of-day is a strong, reliable predictor.

Dynamic pricing already works. The 57 stations already on dynamic pricing run 9.5 points higher utilization than the 190 flat-rate ones.

A natural experiment hiding in the data

The dataset already contains stations with and without dynamic pricing. The gap between them is our motivation.

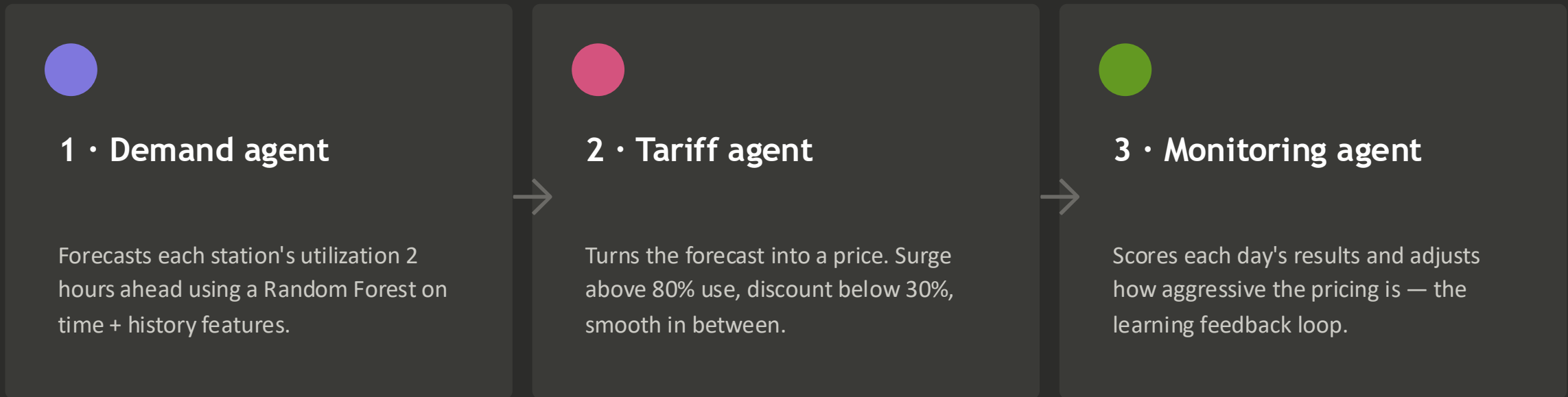


+9.5 pts

higher average utilization on stations that already use dynamic pricing

This is correlation, not proof of causation — but it's a strong signal that demand-aware pricing fills idle capacity.

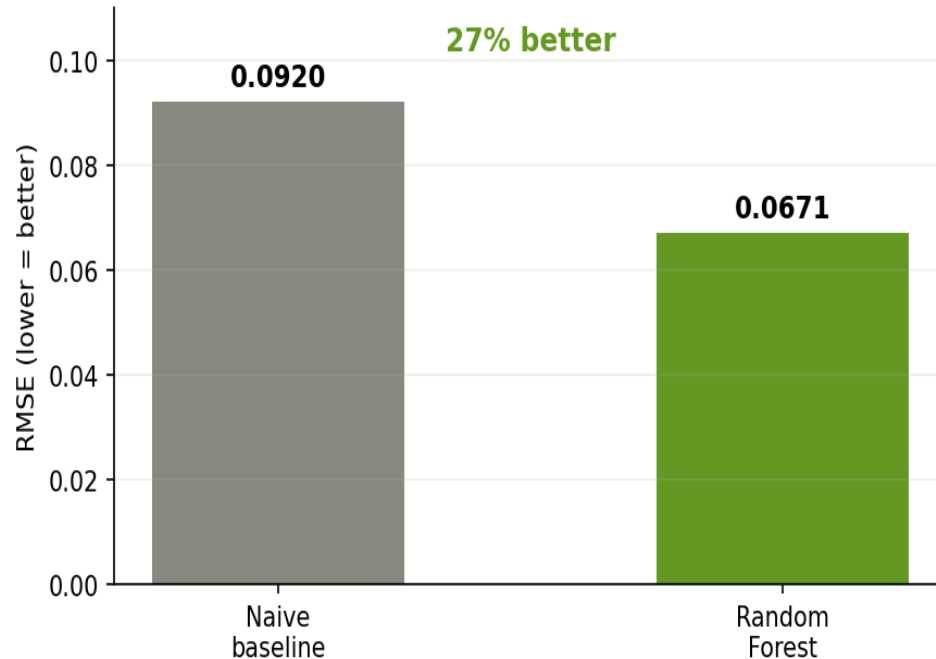
The system: three agents, one loop



The monitoring agent feeds its findings back to the tariff agent — so pricing decisions improve over time.

Demand prediction agent

We predict utilization 2 hours ahead — far enough that the model has to actually learn, not just copy the present.



Why this matters

We tested our model against a **naive baseline** that simply predicts "2 hours from now = right now."

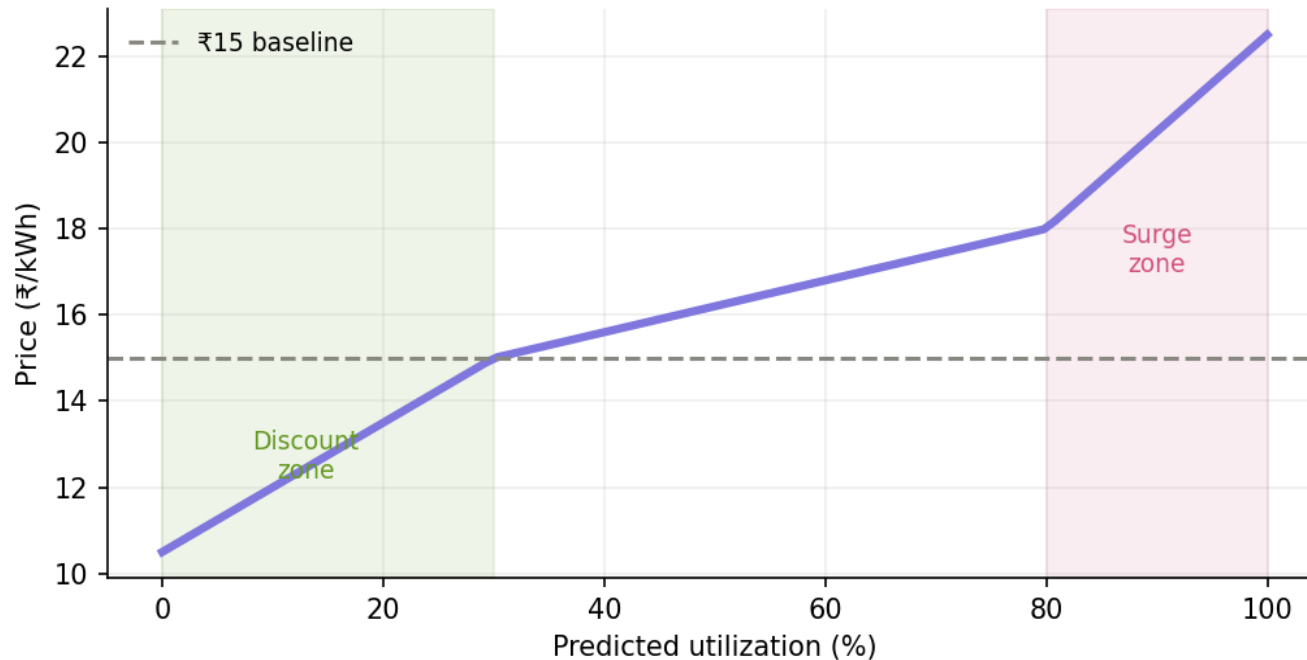
At a 30-minute horizon, both scored $R^2 \approx 0.94$ — the model added nothing, because utilization barely moves in 30 minutes.

At 2 hours, our Random Forest beats the naive baseline by **27% on error**. The engineered features genuinely carry signal — and 2 hours gives operators real lead time to act.

Final scores: $R^2 = 0.86$ · $RMSE = 0.067$ · $MAE = 0.041$

Tariff pricing agent

The forecast becomes a price using simple, explainable rules tied to the 80% / 30% thresholds.



+2.76%

revenue vs ₹15 flat rate

+2.10%

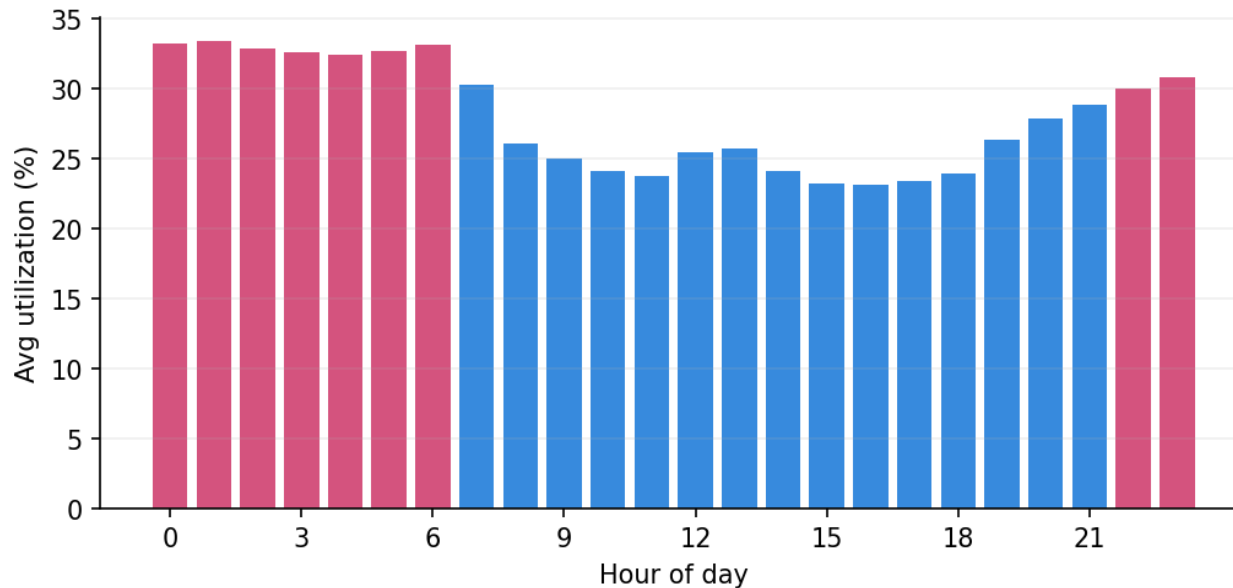
off-peak demand uplift

₹15.62

revenue per kWh (was ₹15.00)

Monitoring & learning agent

Each day is one 'episode'. The agent checks the result and adjusts a boldness factor — bolder when revenue improves, cautious when it dips.



Demand concentrates at night (pink) — exactly where surge pricing eases congestion

What the loop achieves

~7.8% average reduction in peak-period demand pressure (our wait-time proxy)

Tracked metrics each episode:

Revenue gain %, peak wait-time reduction, and pricing efficiency (₹/kWh) — all logged to CSV so the loop's progress is visible.

What it means & where it's limited

Business & policy implications

- Operators earn more without new hardware — just smarter pricing
- Discounts pull demand into empty off-peak windows, easing peak queues
- Per-station pricing respects that CBD and residential areas behave differently
- The feedback loop means the system keeps improving as conditions change

Honest limitations

- Elasticity is assumed (-0.3), not measured from this data
- Wait time is a demand-pressure proxy, not a true queue model
- Shenzhen patterns may not transfer directly to Indian cities
- Revenue gains depend on the elasticity assumption holding true
- We avoid causal claims — the dynamic-vs-flat gap is correlational

Thank you

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